

LocationCircle — Closed-Group Location Sharing Platform

This document presents high-fidelity wireframe mockups for every primary screen of the LocationCircle web / PWA application. Each screen is annotated with component descriptions, UX rationale, accessibility notes, and references to the corresponding requirements in the Business Requirements Document (BRD v1.0).

Screen	BRD Ref	Description
<i>S1 – Splash / Login</i>	FR-AUTH-01, 02	Google sign-in and first launch
<i>S2 – Onboarding</i>	FR-AUTH-02, 03	Profile setup wizard (name, photo, phone)
<i>S3 – Dashboard (Member List)</i>	FR-DASH-01–07, FR-LOC-03	Main landing page with proximity list
<i>S4 – Map View</i>	FR-MAP-01–05	Drill-down map with member pins
<i>S5 – Account Settings</i>	FR-SET-01–05	User profile and preferences
<i>S6 – Group Management</i>	FR-GRP-01–07, FR-SET-02	Head admin tools: invite, temp head, remove
<i>S7 – WhatsApp Bot UI</i>	FR-WA-01–04	In-WhatsApp chat bot interface
<i>S8 – Accessibility Overlays</i>	NFR Accessibility	TTS controls and focus ring demonstration

Design System Reference

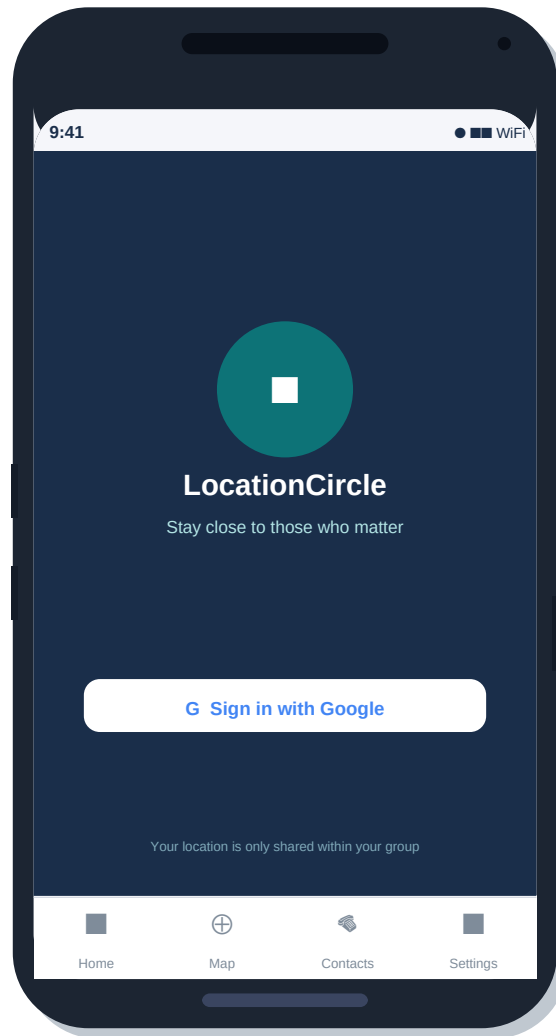
All screens use the following shared design tokens. Colours are chosen to achieve WCAG 2.1 AA contrast ratios (minimum 4.5:1 for body text). The typeface stack uses system-native sans-serif for performance and predictable rendering across Android and iOS.

Token / Value	Usage
Navy #1A2E4A	Primary background, top bars, text headings
Teal #0D7377	Interactive elements, icons, nav active state
Accent #F0A500	Group Head highlight, star badges, CTAs
Green #27AE60	Call button, success states, Temp Head active
Red #E74C3C	Danger actions (delete, remove), alerts
Light Grey #F5F6FA	Screen backgrounds, alternating rows
Body #2C3E50	Primary body text
Font: System Sans	Helvetica/SF Pro/Roboto — OS-native fallback stack
Border radius: 5–10px	Cards, buttons, input fields
Touch target: ≥ 44×44px	All interactive elements (WCAG 2.5.5)

S1 – Splash / Login Screen

■ BRD Ref: FR-AUTH-01, FR-AUTH-02

The splash screen is the first screen a new or signed-out user sees. It communicates the app brand, value proposition, and provides the single sign-on entry point via Google OAuth. No username/password is accepted — Google is the only identity provider (FR-AUTH-01).



Login

Screen Annotations

Logo + Brand	Circular teal logo with pin icon, app name in white, tagline in muted teal — sets tone and purpose immediately.
Tagline	'Stay close to those who matter' — reinforces the emotional value proposition; low cognitive load.
Google Sign-In	Single prominent white button with Google branding. OAuth 2.0 flow opens in-browser; no credentials stored by the app. (FR-AUTH-01)
Privacy Note	Micro-copy below button: 'Your location is only shared within your group' — pre-empts privacy anxiety.
No Nav Bar	Navigation and status bar are hidden on this screen; full focus on sign-in action.

Accessibility Notes

- Button has visible focus ring (3px teal outline) for keyboard / switch-access users.
- Google button text includes 'G' glyph — not purely icon-based — ensuring screen readers read 'Sign in with Google'.
- Minimum contrast 4.5:1 on tagline text against dark background (verified).
- No CAPTCHA or additional barriers — OAuth handles bot prevention.

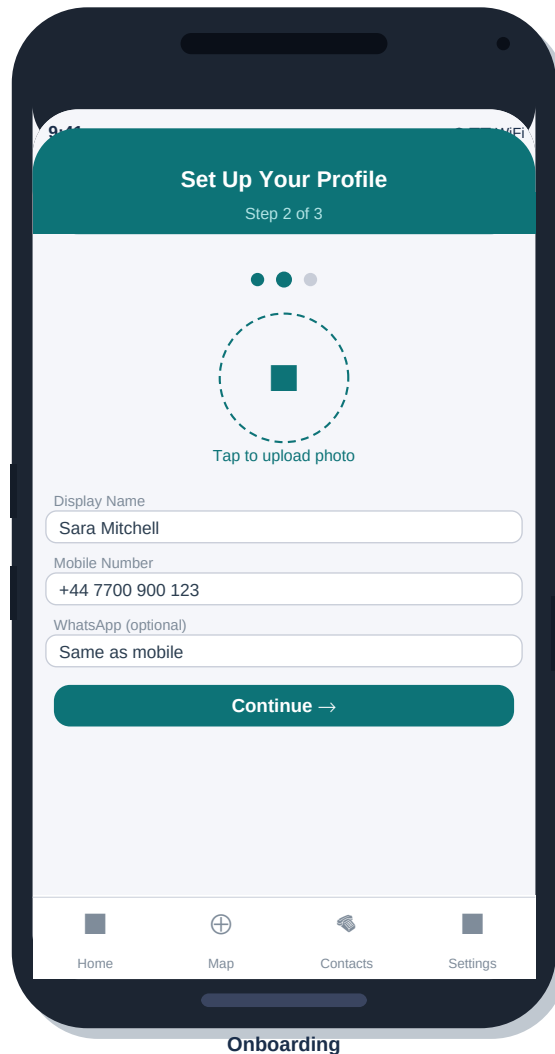
UX Rationale

- Single CTA eliminates decision paralysis. Google OAuth reduces signup friction to two taps.
- Dark branded background creates strong first impression and sets visual hierarchy for the rest of the app.

S2 – Onboarding – Profile Setup

■ BRD Ref: FR-AUTH-02, FR-AUTH-03

Shown only on first login. A three-step wizard collects the information required to make the member list useful: display name, profile photo (used as map pin), and phone number (required for the Call feature). WhatsApp linking is optional at this stage (FR-AUTH-03) and can be completed later in Account Settings.



Screen Annotations

Progress Dots	Three dots indicate wizard step (step 2 of 3 shown). Gives users a sense of progress and finite commitment.
Photo Upload	Dashed circle invites tap-to-upload. Profile photo becomes the map marker icon (FR-MAP-02). Falls back to initials avatar if skipped.
Display Name	Pre-populated from Google account name; editable. Used throughout the app and on the WhatsApp bot. (FR-AUTH-02)
Mobile Number	Required for the Call button (FR-DASH-04) and WhatsApp OTP (FR-AUTH-03). E.164 format validated on blur.

WhatsApp Field	Optional — defaults to 'Same as mobile'. Triggers OTP verification on Continue if different. (FR-AUTH-03)
Continue Button	Full-width teal CTA. Disabled until Name and Phone are valid. Shows inline error states on invalid submit.

Accessibility Notes

- All form inputs have visible labels above the field (never placeholder-only) for screen-reader compatibility.
- Error messages use colour + icon + text — not colour alone — for colour-blind users.
- Step progress communicated via aria-label='Step 2 of 3' on the progress indicator.
- Photo upload is optional; avatar initials fallback ensures no broken UI for screen readers.

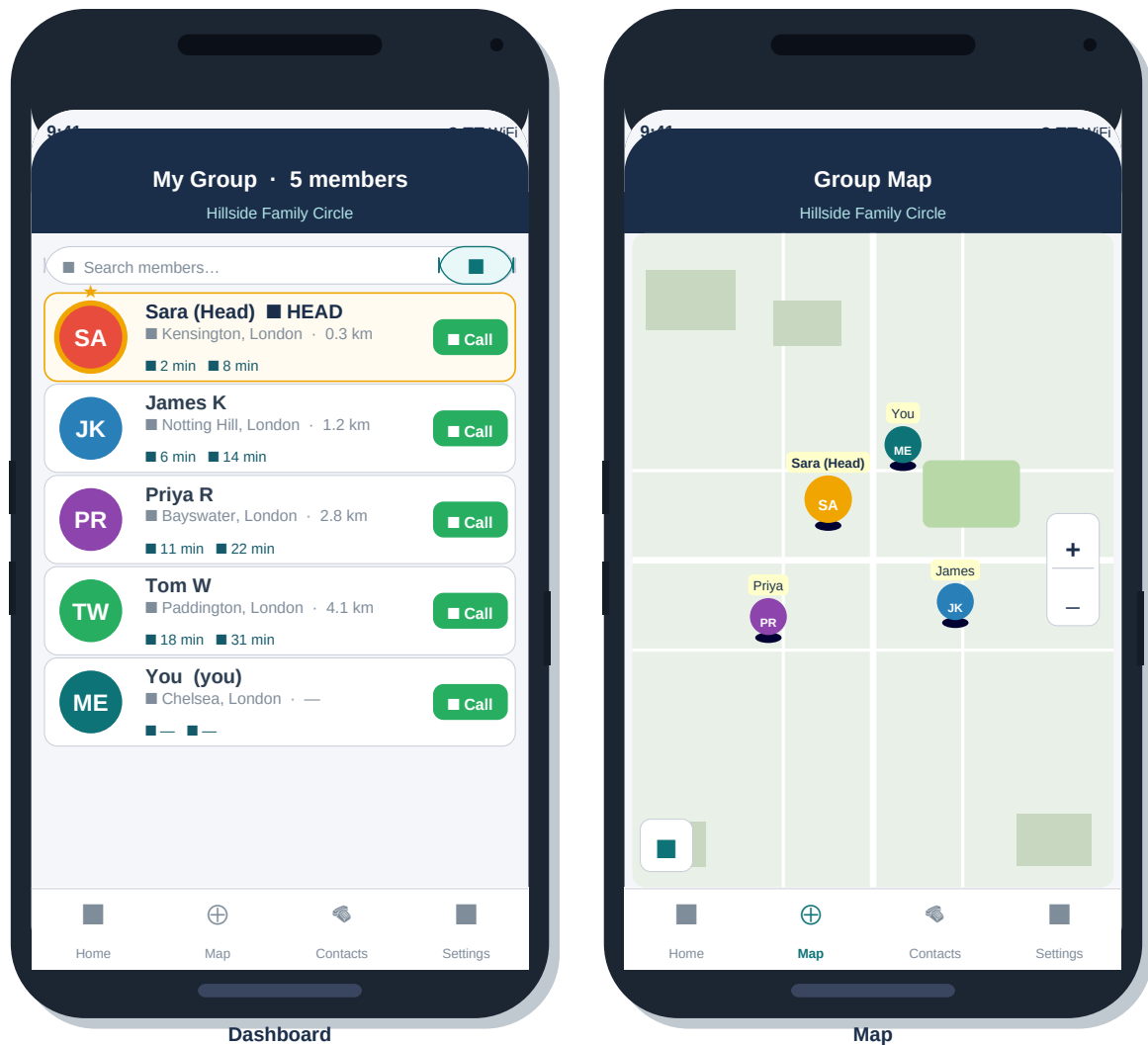
UX Rationale

- Wizard pattern limits cognitive load — users see only the fields needed per step.
- Making WhatsApp optional here reduces drop-off; users can link later when they see the value.

S3 – Dashboard (Member List)

■ BRD Ref: FR-DASH-01-07, FR-LOC-01-06, FR-AUTH-04

The main landing screen after login. A scrollable, proximity-sorted list of all group members showing real-time locality, distance, and ETA by car and public transport. The Group Head is pinned to the top with a gold highlight and crown badge. A Read Aloud button enables TTS playback of the entire list (FR-DASH-06), supporting low-vision users.



Dashboard Annotations (S3)

App Bar	Shows group name and member count. Group Head name/status displayed as subtitle. Taps navigate to Group Management. (FR-DASH-01)
Search Bar	Filters member list in real-time as user types. Useful in larger groups. Keyboard-accessible with clear button.
■ Read Aloud	Activates Web Speech API TTS. Reads each member row: Name, Location, Distance, ETA. Pause button appears during playback. (FR-DASH-06)
★ HEAD Row	Gold border, crown icon, pinned top position distinguish the Group Head (or active Temp Head). Updated immediately on head change. (FR-DASH-02, FR-LOC-02)

Member Row	<i>Profile avatar · Name · Locality (closest town) · Distance · ETA car & transit. Tapping drills to map. (FR-DASH-03, FR-LOC-03)</i>
■ Call Btn	<i>Green button triggers tel: URI intent → native phone dialler. Accessible label: 'Call [Name]'. (FR-DASH-04)</i>
(you) Tag	<i>Viewer's own row tagged '(you)'; Call button hidden; distance shows '—'. Location pause toggle accessible here.</i>
Nav Bar	<i>Home / Map / Contacts / Settings. Active tab highlighted in teal with bold icon.</i>

Map View Annotations (S4)

■ BRD Ref: FR-MAP-01–05

Map Provider	<i>Google Maps JavaScript SDK (or Mapbox GL JS fallback). Satellite / street / hybrid toggle via long-press on map. (FR-MAP-04)</i>
Member Pins	<i>Circular profile photo inside a teardrop pin shape. Head pin is gold with larger radius. All pins show name label on tap. (FR-MAP-02)</i>
'You' Pin	<i>Teal 'ME' pin with pulsing ring animation — visually distinct from other members. (FR-MAP-03)</i>
Zoom Controls	<i>+ / – buttons in bottom-right for users who cannot pinch-zoom (motor accessibility). (FR-MAP-04)</i>
■ Locate Me	<i>Bottom-left button re-centres map on user's current position.</i>
ARIA Region	<i>Hidden aria-live region announces 'Sara is now in Kensington' when locations update. (FR-MAP-05)</i>

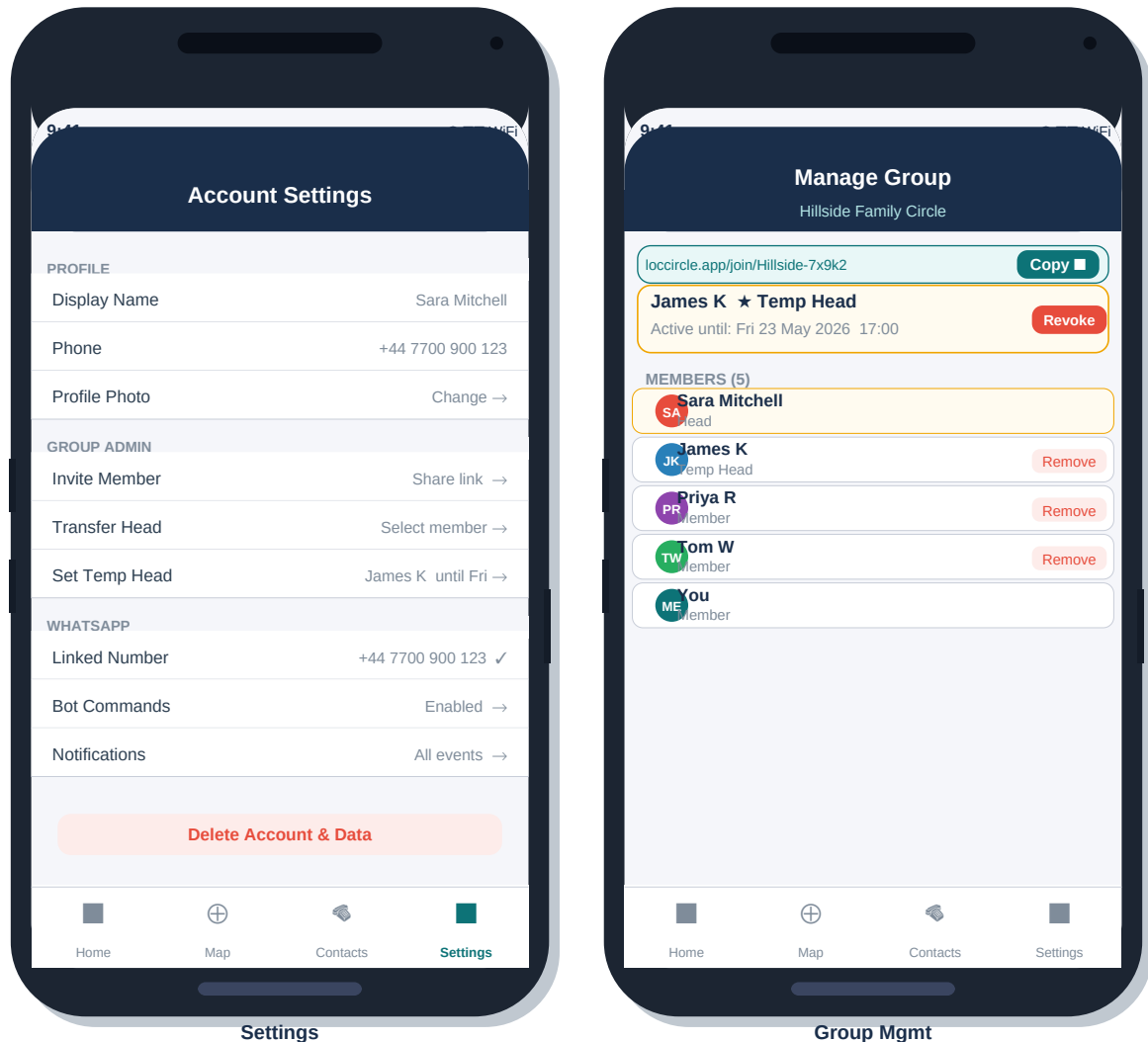
Accessibility Notes — Dashboard & Map

- Each member row is a element with role='listitem'; aria-label combines all visible text for coherent TTS read.
- Distance and ETA values use aria-live='polite' so screen readers announce updates without interrupting.
- Call button aria-label='Call Sara Mitchell' — not just 'Call' — provides unique labelling per row.
- Map pins include aria-label='Sara Mitchell, Kensington, 1.2 km away, Head' read by screen readers.
- Colour is never the sole differentiator: Head row uses colour + icon + position + border.
- Pinch-zoom supplemented by +/– buttons for motor-impaired users.

S5 – Account Settings & S6 – Group Management

■ BRD Ref: FR-SET-01-05, FR-GRP-01-07

Account Settings allows all members to manage their profile, phone, and WhatsApp link. Group Management (accessible only to the Group Head) provides member invitation, permanent head transfer, and Temporary Head assignment with expiry date/time. Both screens are accessible via the ■ Settings tab in the bottom nav bar.



Settings Screen Annotations (S5)

Profile Section	Display name, phone, and profile photo — all editable inline with tap. Changes sync across all group members immediately. (FR-SET-01)
Group Admin	Visible only to Group Head. Provides Invite, Transfer Head, and Set Temporary Head controls. Hidden from regular members. (FR-SET-02)
WhatsApp Section	Shows linked number with ✓ verification badge. Toggle bot commands on/off. Notification preferences per event type. (FR-SET-03, FR-SET-04)
Temp Head Row	Shows current Temp Head name and expiry. 'Revoke' button cancels delegation immediately. Countdown shown in hours if < 24h remain.

Delete Account	<i>Red danger zone — requires two-step confirmation ('Type DELETE to confirm'). Triggers GDPR erasure pipeline. (FR-SET-05)</i>
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Group Management Annotations (S6)

Invite Link	<i>Auto-generated shareable URL. Copy button. Link expires after 7 days or after 1 use — configurable by head. (FR-GRP-02)</i>
Temp Head Card	<i>Gold-bordered card showing active Temp Head, expiry date/time, and Revoke button. Only one Temp Head at a time. (FR-GRP-04)</i>
Member List	<i>All members listed with role badge. Head cannot remove themselves. Remove button triggers confirmation dialog. (FR-GRP-06)</i>
Transfer Head	<i>Tap member row → action sheet: 'Make permanent Head' / 'Set as Temp Head'. Permanent transfer immediately demotes current Head to member. (FR-GRP-03)</i>
Temp Head Expiry	<i>Date/time picker. On expiry, server-side cron reinstates original Head and sends WhatsApp + push notification to group. (FR-GRP-04)</i>

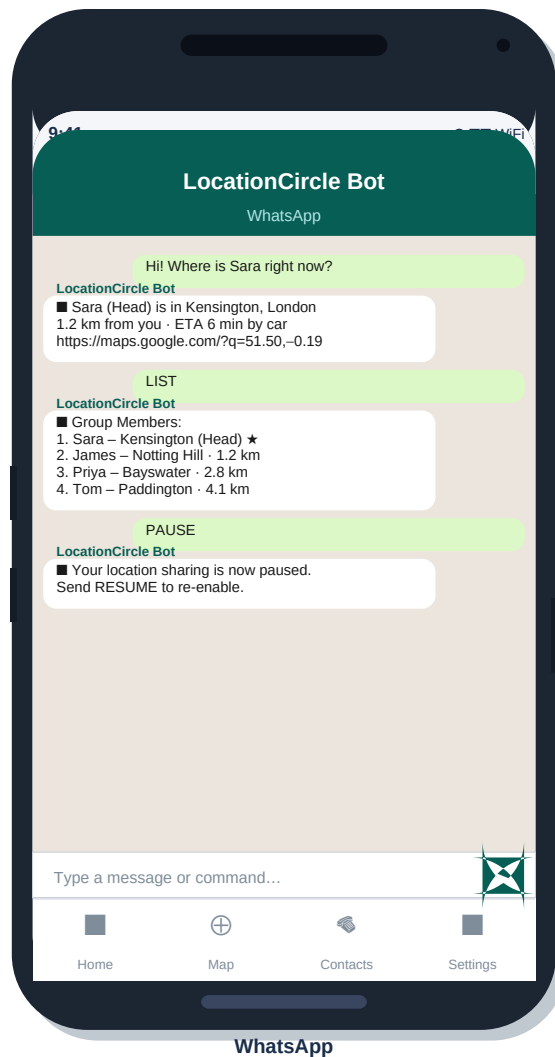
Accessibility Notes

- Settings rows use native form controls wrapped in for full screen-reader compatibility.
- Destructive actions (Remove, Delete, Revoke) require explicit confirmation — prevents accidental activation.
- Danger zone button uses red colour + text description — never icon-only.
- Role badges ('Head', 'Temp Head', 'Member') use aria-label on list items for TTS context.

S7 – WhatsApp Bot Interface

■ BRD Ref: FR-WA-01-04, FR-AUTH-03

Members who prefer not to open the app can interact entirely through the official WhatsApp Business API bot. The bot responds to natural-language queries and structured commands. Responses include deep-link URLs back to the app for users who wish to switch to the full interface. The screen shown represents the bot as seen inside WhatsApp — not a view inside the LocationCircle app itself.



Screen Annotations

Bot Header	Standard WhatsApp chat header. Bot display name 'LocationCircle Bot' with green verified badge.
Natural Query	'Where is Sara right now?' — the bot interprets intent and responds with locality + Google Maps link. (FR-WA-02)
LIST Command	Returns all members sorted by proximity with head badge ★. Formatted for plain text. (FR-WA-03)
PAUSE / RESUME	Members can pause and resume their own location sharing from WhatsApp. Confirmation message sent. (FR-WA-03)
HEAD Command	Returns the Group Head's (or Temp Head's) current position and ETA from user. Not shown in this view.

Deep Link in Reply	<i>Each location response includes a Google Maps link and an app deep-link for map drill-down.</i>
Input Bar	<i>Standard WA input; bot also accepts free-text queries in addition to commands.</i>

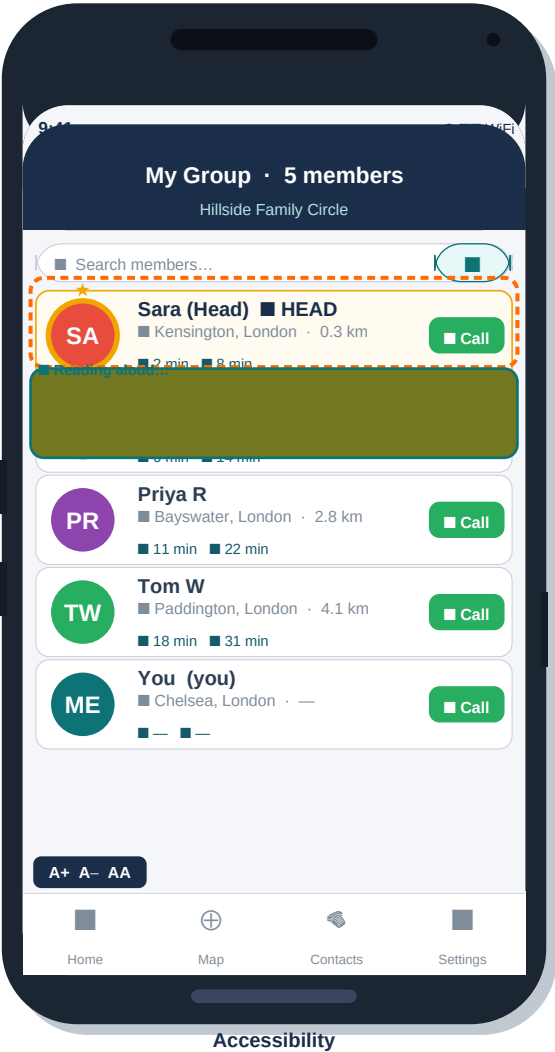
Accessibility Notes

- All bot messages use plain text + emoji — no images or attachments that might fail on older handsets.
- Bot responses tagged with sender name ('LocationCircle Bot') for screen-reader disambiguation.
- Commands are single English words (LIST, HEAD, PAUSE, RESUME) — minimal typing burden. (FR-WA-03)
- Location data in messages is text-first (locality + distance) with map link secondary — accessible without clicking. (FR-WA-04)
- Bot respects GDPR: no member locations stored in WhatsApp; always fetched live from API.

S8 – Accessibility Feature Highlights

■ BRD Ref: NFR – Accessibility (WCAG 2.1 AA), FR-DASH-05, FR-DASH-06

This section illustrates the accessibility features overlaid on the Dashboard screen. Key features include visible focus rings, TTS Read Aloud mode, font scaling, and high-contrast mode. All features are available without any external assistive technology — they are built into the app UI itself.



Accessibility Feature Reference

Focus Ring (orange)	3px solid orange focus outline on keyboard-focused element. Visible at all zoom levels. Meets WCAG 2.4.7 Focus Visible (AA).
TTS Active Row	Teal highlight + 'Reading aloud...' label shows which row is being spoken. TTS reads: Name, Location, Distance, ETA, then 'Call button'. (FR-DASH-06)
A+ A- Font Scale	In-app font size controls. Layout reflows at 200% without horizontal scroll (WCAG 1.4.4 Resize Text).
AA Contrast Badge	WCAG 2.1 AA badge shown in dev/QA mode. All text combinations verified $\geq 4.5:1$ normal, $\geq 3:1$ large.
Keyboard Navigation	Tab order: App Bar → Search → TTS Button → Member Rows (top to bottom) → Call buttons → Nav Bar. Arrow keys scroll list.

Skip Link	Hidden 'Skip to member list' link appears on first Tab keypress — jumps past nav for keyboard users.
ARIA Live Regions	Location updates announced via <code>aria-live='polite'</code> ; urgent alerts (Head changed) via <code>aria-live='assertive'</code> .
Colour Independence	Head row uses position (pinned) + icon (★) + border + colour — 4 independent cues. Passes for all colour-blindness types.

User Flow Summary

Step	Screen	Trigger / Action	Next
1	S1 – Login	App first launch / signed out	Google OAuth success → S2 (new) or S3 (returning)
2	S2 – Onboarding	First login only	Complete wizard → S3 Dashboard
3	S3 – Dashboard	Default post-login screen	Tap member → S4 Map; tap Settings → S5
4	S4 – Map	Tap any member row on Dashboard	Back → S3; tap pin callout → member detail sheet
5	S5 – Settings	Tap Settings nav	Head sees Group Admin section → S6 Group Mgmt
6	S6 – Group Mgmt	Head taps Manage Group in Settings	Invite / Temp Head / Remove → back to S5
7	S7 – WhatsApp	User opens WhatsApp, messages bot	Bot replies; deep-link back to S3 or S4
8	S3 – Dashboard	Real-time location update received	List refreshes in place; ARIA live region fires